

SPHINX V2 – PROGRAMMABLE CORE POWER

Core: 0.75 V nominal, 5 W active, 1 GHz target. Range: 0.30–1.20 V. I/O: 1.8 V. Schematic for PCB layout.

Protected input and auxiliary supply

01_input_aux.kicad_sch

File: 01_input_aux.kicad_sch

PMBus core regulator, 0.30–1.20 V

02_core_regulator.kicad_sch

File: 02_core_regulator.kicad_sch

Fixed 1.8 V I/O and calibrated telemetry

03_io_and_telemetry.kicad_sch

File: 03_io_and_telemetry.kicad_sch

USB control and remote reset

04_usb_control.kicad_sch

File: 04_usb_control.kicad_sch

Hardware power shutdown latch

05_hardware_safety.kicad_sch

File: 05_hardware_safety.kicad_sch

Socket pinout and signal interfaces

06_socket_and_signals.kicad_sch

File: 06_socket_and_signals.kicad_sch

Distributed core decoupling and test access

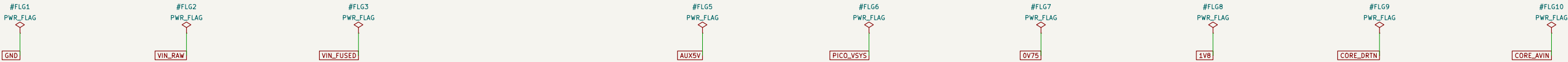
07_core_decoupling.kicad_sch

File: 07_core_decoupling.kicad_sch

I/O decoupling and voltage monitor

08_io_decoupling.kicad_sch

File: 08_io_decoupling.kicad_sch



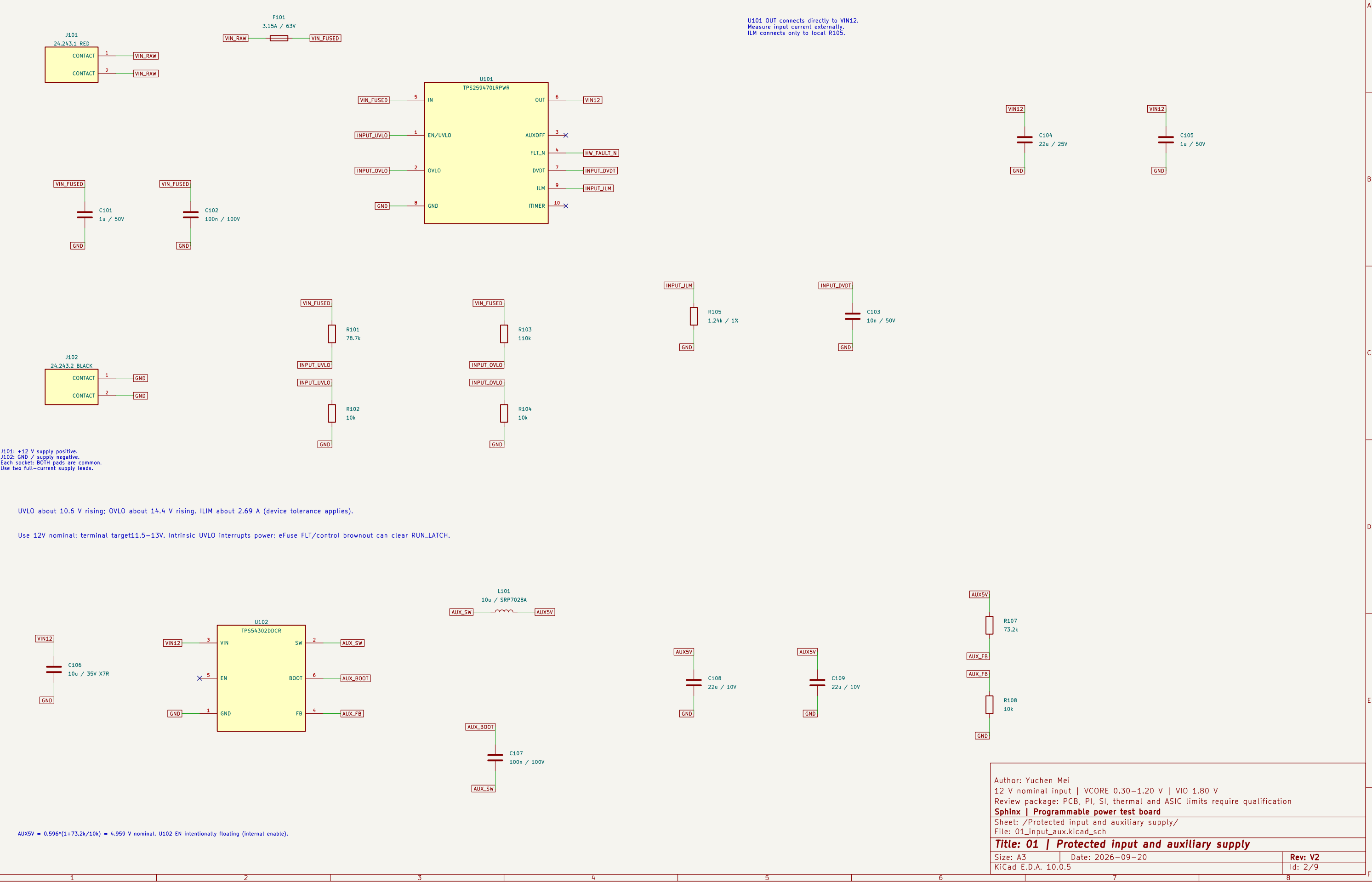
12 V input, eFuse, PMBus core regulator, current shunt, socket VCore; fixed 1.8 V I/O. Core/I/O telemetry; Pico USB control; 12+12 capacitor population; hardware power shutdown latch and STOP. Valid VIO: reset defaults released; explicit button/ZedBoard/Pico reset only. PG and VIN12_VALID are monitors.

10 A continuous design target. Characterize voltage, frequency, power, and temperature per DUT. Change feedback scale with outputs disabled. Layout requirements: docs/LAYOUT_HANDOFF.md.

Author: Yuchen Mei		
12 V nominal input VCore 0.30–1.20 V VIO 1.80 V		
Review package: PCB, PI, SI, thermal and ASIC limits require qualification		
Sphinx Programmable power test board		
Sheet: /		
File: sphinx_power_v2.kicad_sch		
Title: Sphinx V2 Programmable power test platform		
Size: A3	Date: 2026–09–20	Rev: V2
KiCad E.D.A. 10.0.5		Id: 1/9

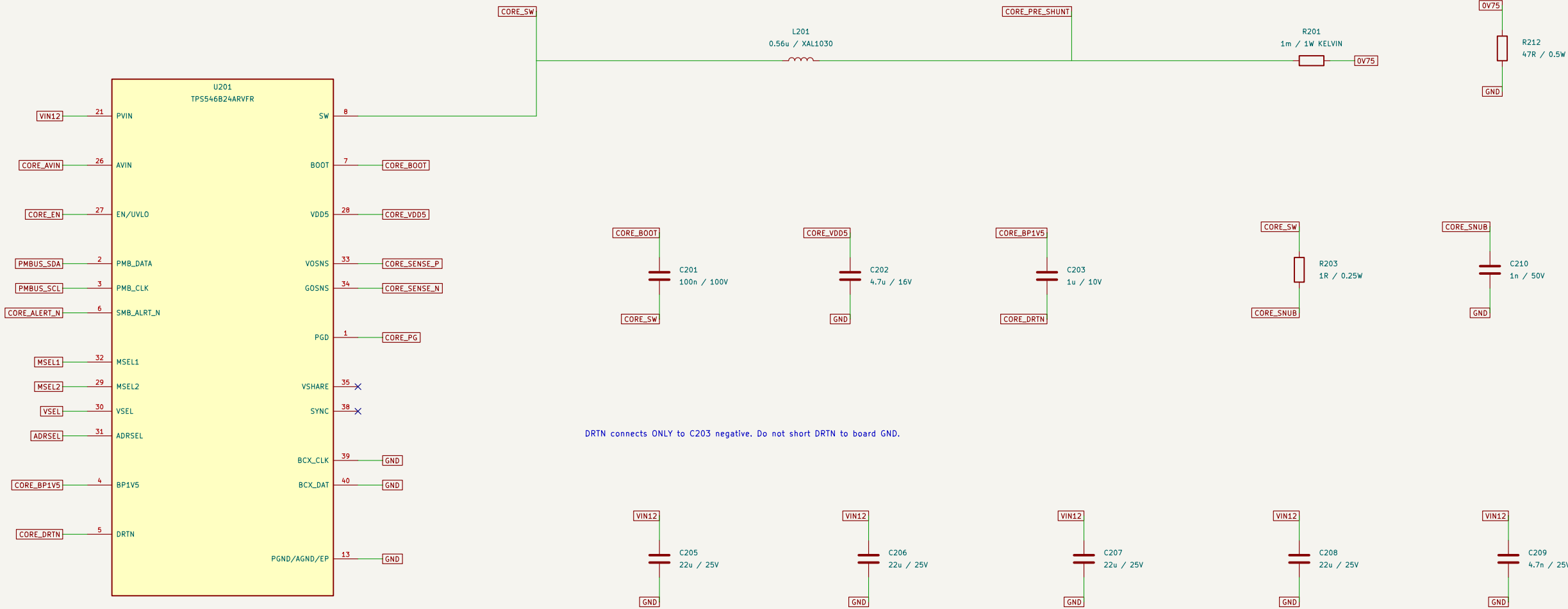
INPUT: use 12 V / 3 A bench supply. OVLO disconnects a steady 20 V mis-setting.

F101 is backup protection. U101 UVLO/OVLO auto-recover; U509 VIN12_VALID is monitor-only.



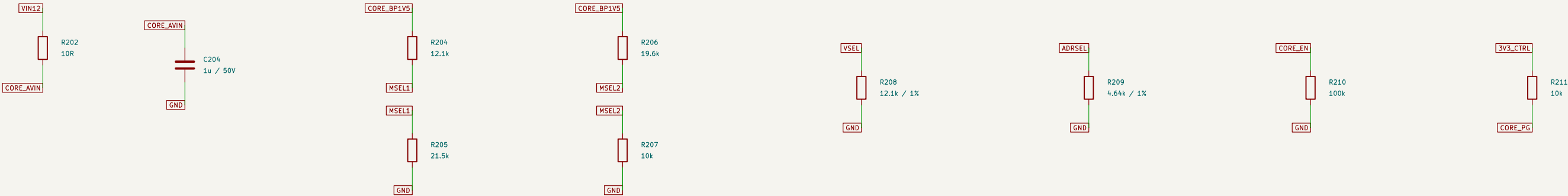
0.75 V nominal core; approximately 5 W during kernel execution. Board continuous-current design target: 10 A.

Boot 0.750 V, 650 kHz. Differential sense after R201 at socket. Output remains disabled until hardware run latch + MCU enable.



DRTN connects ONLY to C203 negative. Do not short DRTN to board GND.

Select 12+12 or 16+16 population; program and read back B1 with output OFF. Feedback scales: 1 and 0.5.



PMBus 7-bit address: 0x10, R209 = 4.64k to GND, no top resistor; SYNC auto-detect. Firmware must match.

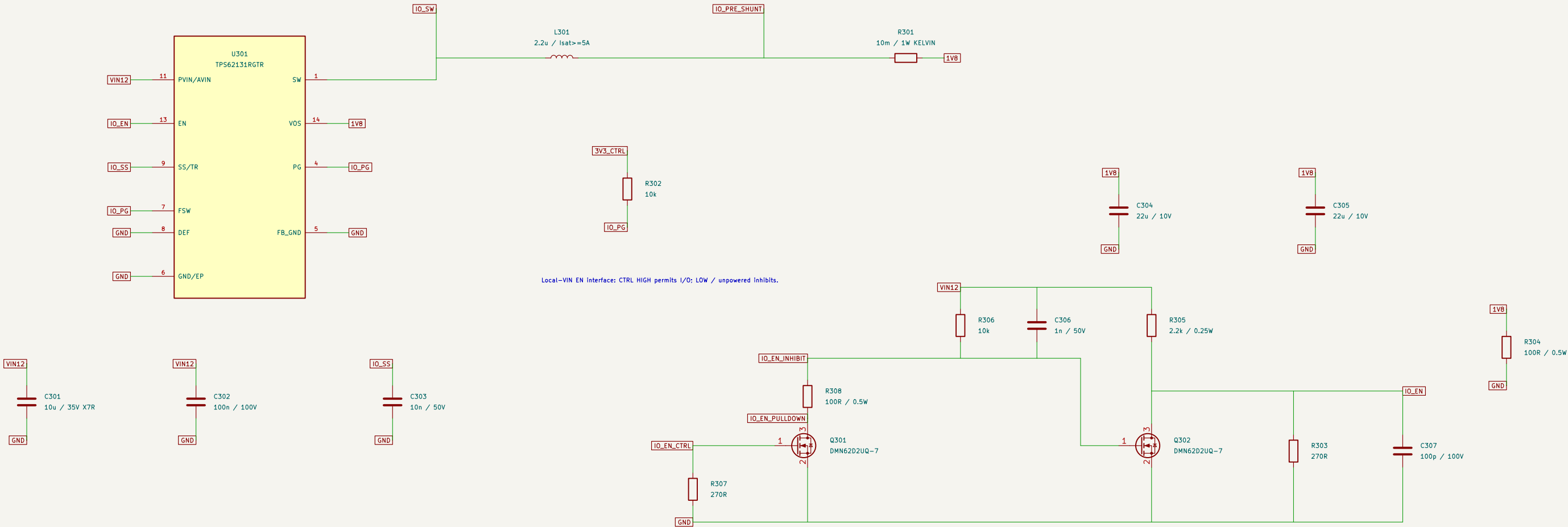
BCX_CLK/DAT and NC pin36 grounded for stand-alone operation, AGND37 and all PGND/EP pads share local return.

Author: Yuchen Mei
12 V nominal input | V_{CORE} 0.30–1.20 V | V_{IO} 1.80 V
Review package: PCB, Pl, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /PMBus core regulator, 0.30–1.20 V/
File: 02_core_regulator.kicad_sch

Title: 02 | PMBus core regulator, 0.30–1.20 V

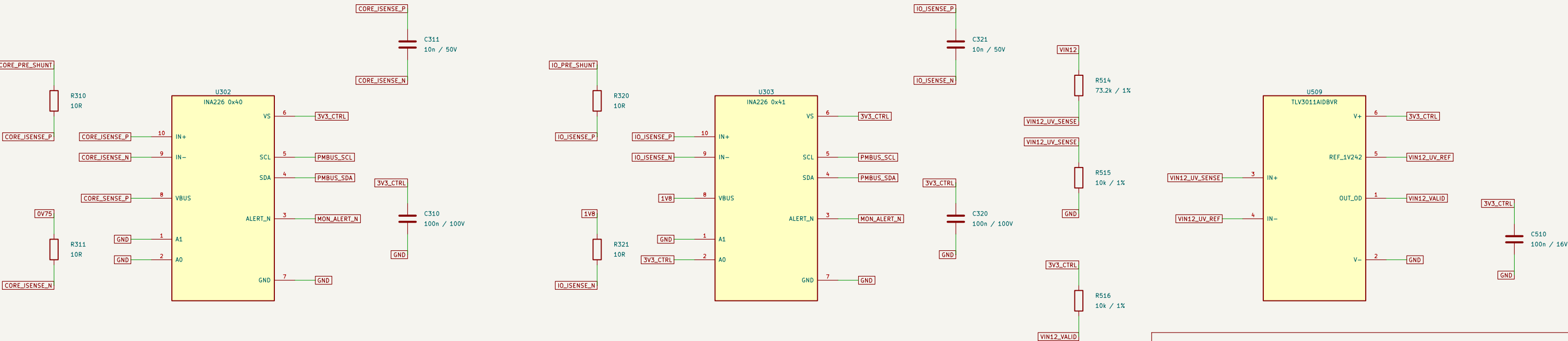
Size: A3	Date: 2026-09-19	Rev: V2
KiCad E.D.A. 10.0.5		Id: 3/9

VIO: fixed 1.80 V / 3 A regulator capability; firmware default load budget 1 A. No I/O voltage sweep.



Local-VIN EN interface: CTRL HIGH permits I/O: LOW / unpowered inhibits.

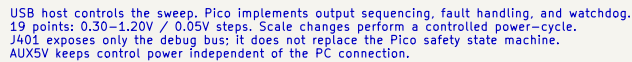
FSW tied to PG selects about 1.25MHz after startup. PG may float HIGH while OFF; U508 IQ_VALID monitors VIO independently.



U509: VIN12_VALID to Pico GP12, monitor-only; nominal threshold10.33V. Intrinsic UVLO or control brownout can still interrupt rails.

INA226 averaging window2.656ms: average rail telemetry only. No input-current ADC; bench supply reads total input.

USB commands: SET / ON / OFF / RESET / STATUS. Hardware controls: STOP and DUT RESET; reset is explicit only.



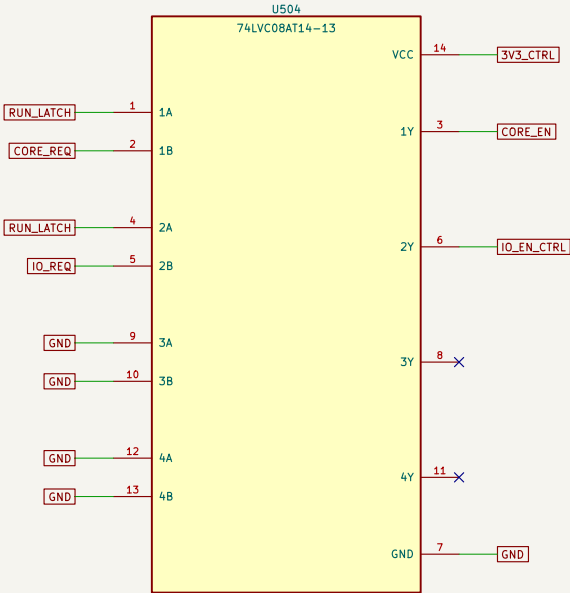
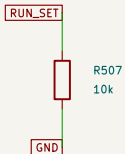
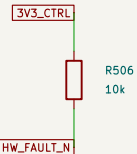
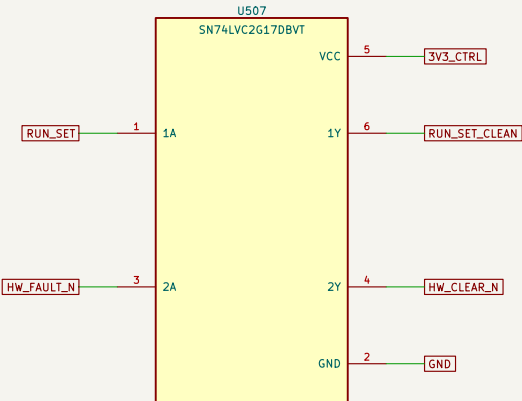
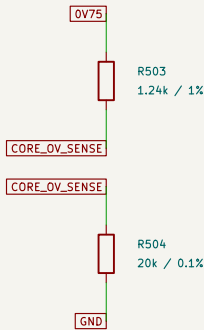
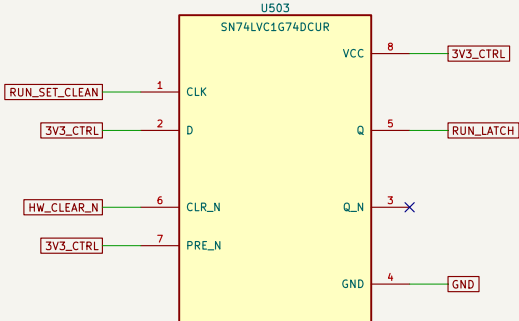
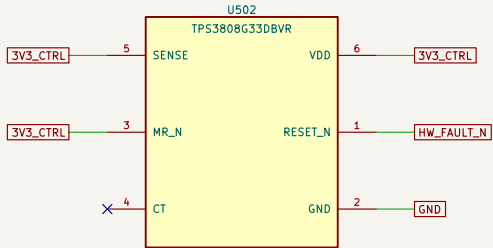
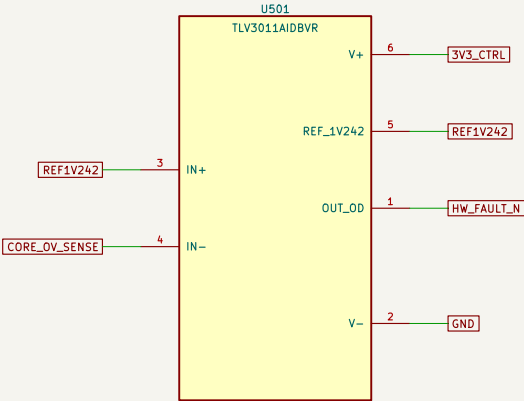
GP15: OPEN-DRAIN shutdown; LOW clears RUN_LATCH. Never drive HW_FAULT_N push-pull HIGH.

Title: 04 | USB control and remote reset

Size: A3	Date: 2026-09-20	Rev: V2
KiCad E.D.A. 10.0.5		Id: 5/9

ON sets the run latch. STOP, damage–risk faults or control brownout clear it; recovery requires a new ON command.

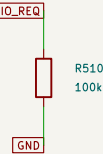
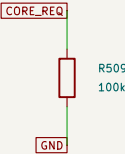
Core OVP and eFuse FLT retain hardware shutdown. VIN12_VALID, CORE_PG and IO_VALID are monitor–only.



GP10: one RUN_SET pulse inside ON, after configuration.
GP15: LOW / high impedance only on HW_FAULT_N.
A stuck–high RUN_SET cannot restart a cleared latch.

DUT reset is independent of RUN_LATCH and PG.
At valid 1.8V, reset defaults LOW (released).
Only SW601, U601 or Pico RESET_ASSERT assert reset.

C502–U501: C503–U502: C504–U503.
C506–U504: C508–U507.
Place each bypass capacitor at its IC.



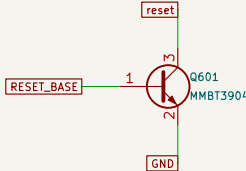
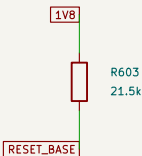
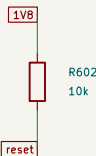
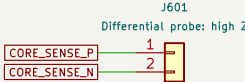
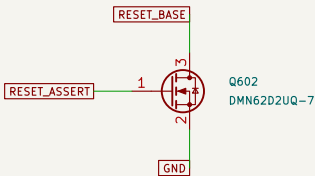
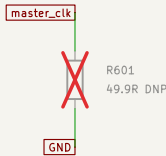
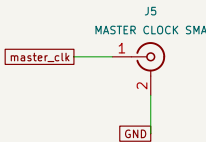
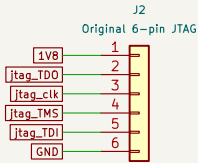
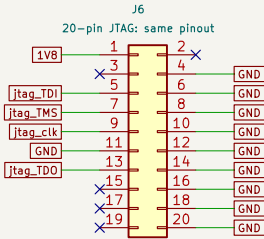
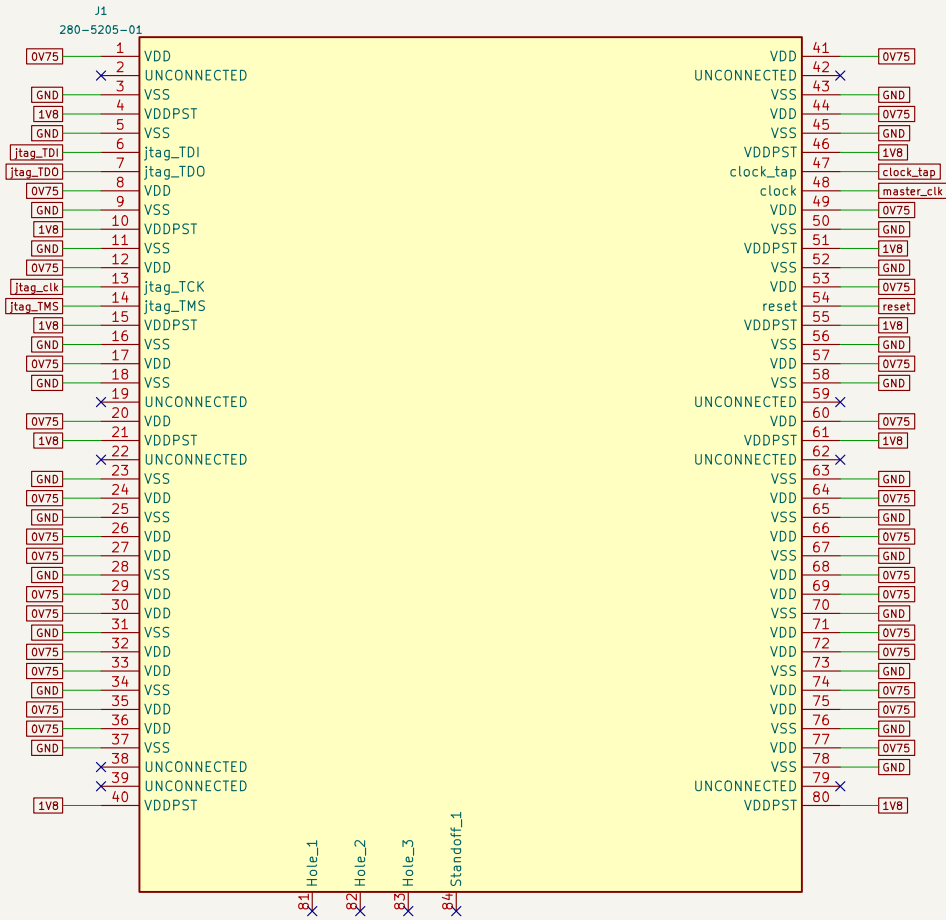
Author: Yuchen Mei
12 V nominal input | VCORE 0.30–1.20 V | VIO 1.80 V
Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /Hardware power shutdown latch/
File: 05_hardware_safety.kicad_sch

Title: 05 | Hardware power shutdown latch

Size: A3 Date: 2026–09–20 Rev: V2
KiCad E.D.A. 10.0.5 Id: 6/9

J1: QFN80 socket, 29 VDD, 25 VSS, 10 VDDPST, 7 signals, and 9 NC positions.

Net 0V75 carries adjustable VCORE. NC positions and the center thermal probe have no electrical connection.



TP7 + TP607: by J1.54, <=2mm pad centres.
High-Z probe; no 50ohm termination.

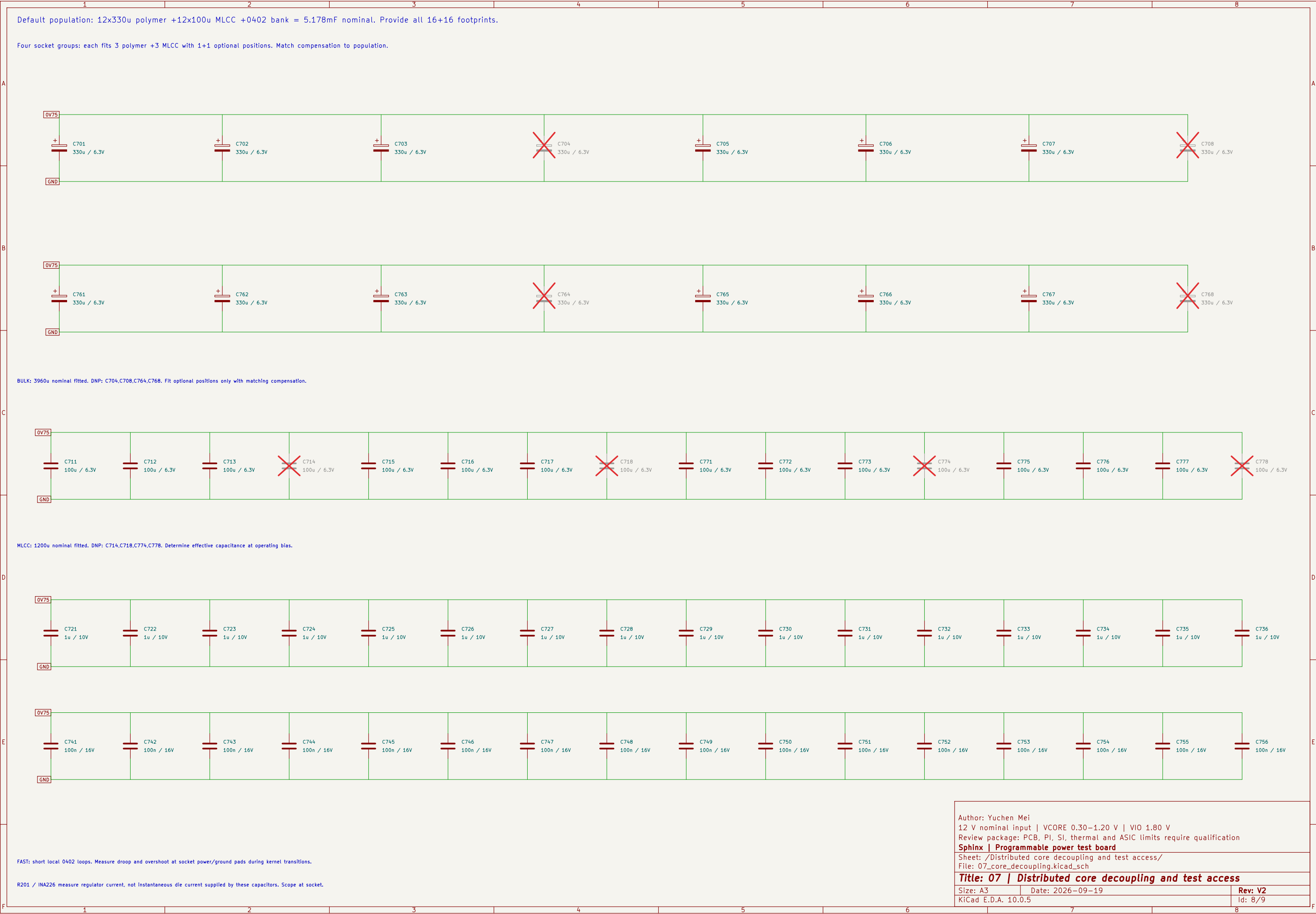
Valid VIO: reset LOW by default; any explicit request asserts HIGH.
GP4 LOW / high-Z releases its request. Place R411 at Q602 gate.

JTAG and clock source must use 1.8V logic and be high-Z while VIO is off. Do not connect a50ohm scope input across VCORE.

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12 V nominal input | VCORE 0.30–1.20 V | VIO 1.80 V
Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /Socket pinout and signal interfaces/
File: 06_socket_and_signals.kicad_sch

Title: 06 | Socket pinout and signal interfaces

Size: A3 Date: 2026-09-20 Rev: V2
KiCad E.D.A. 10.0.5 Id: 7/9



U508 measures VIO directly. IO_VALID reports to Pico GP7 only; regulator IO_PG is only used by FSW.



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12 V nominal input | VCore 0.30~1.20 V | VIO 1.80 V
Review package: PCB, PI, SI, thermal and ASIC limits require qualification
Sphinx | Programmable power test board
Sheet: /I/O decoupling and voltage monitor/
File: 08_io_decoupling.kicad_sch

Size: A3	Date: 2026-09-20	Rev: V2
KiCad E.D.A. 10.0.5		Id: 9/9